# TIC 257773097

Eclipsing binary candidate dossier - Star Vetter

**PERIOD ALIAS**

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BLS EPHEMERIS
  Period       4.945229 d  (ALIASED, see below)
    -> corrected period: 9.890459 d
  Epoch (t0)   1632.3224 BTJD
  Duration     0.1200 d
  Depth        15.891 %  (158,913 ppm)
  SNR          1882.7

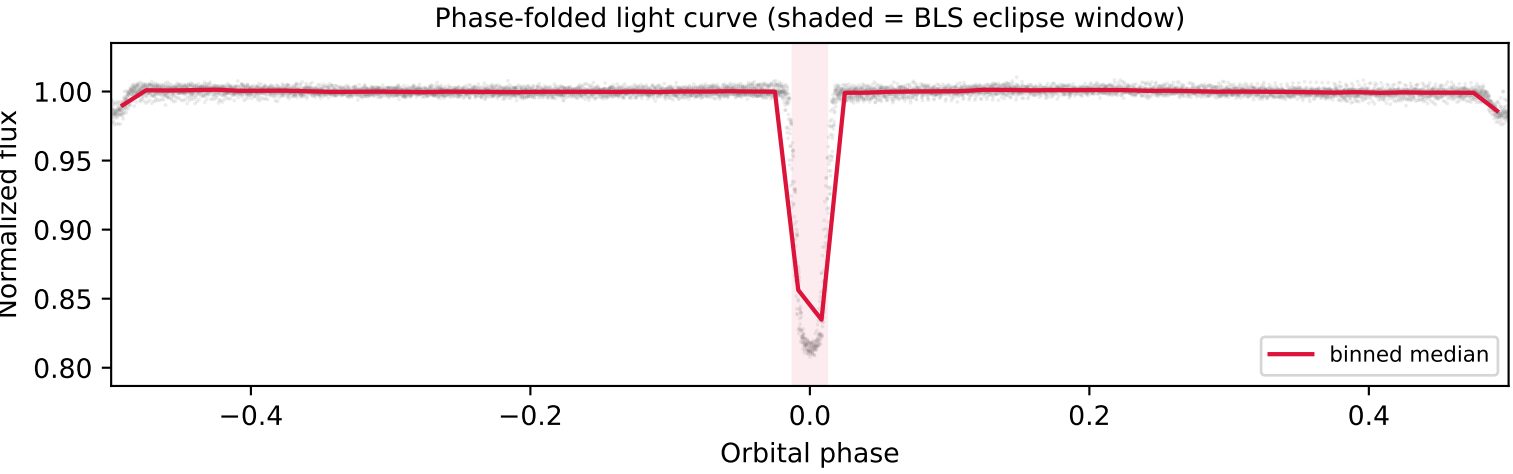
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     20  (gate >= 3)
  Robust depth ratio    1.11  (gate >= 0.50)
  Robust depth sigma    866.8  (gate >= 7.0)
  Odd/even z-score      5.40  (ALIAS FLAGGED)
  Secondary eclipse     detected, phase 0.49, 7.0 sigma
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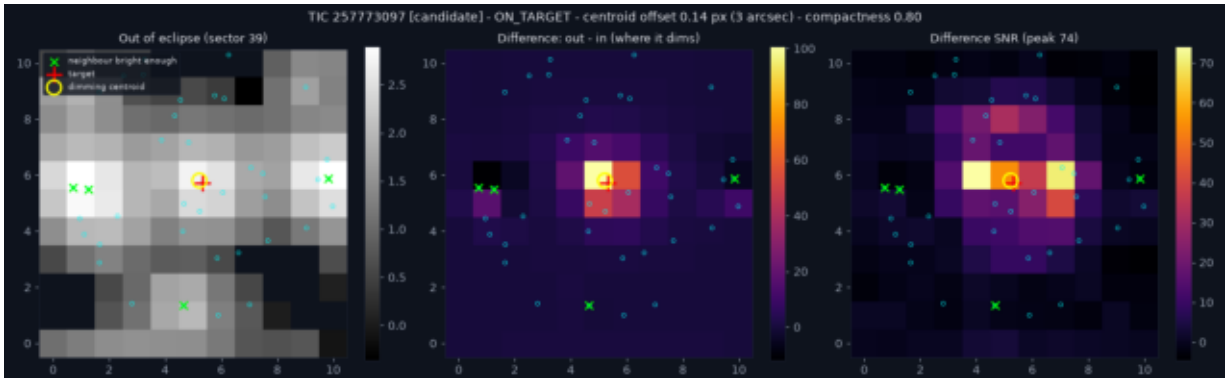
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 39    light-curve dip in this sector: 470.3 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   74.17   (gate >= 5.0)      Compactness   0.80   (gate >= 0.50)
  Dimming centroid offset     0.14 px (3 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 257773109
    3.98 px away, dTmag -0.40 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 5.40 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 9.890459 d, in any write-up.